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Attention-Augmented Machine Memory

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Abstract
Attention mechanism plays an important role in the perception and cognition of human beings. Among others, many machine learning models have been developed to memorize the sequential data, such as the Long Short-Term Memory (LSTM) network and its extensions. However, due to lack of the attention mechanism, they cannot pay special attention to the important parts of the sequences. In this paper, we present a novel machine learning method called attention-augmented machine memory (AAMM). It seamlessly integrates the attention mechanism into the memory cell of LSTM. As a result, it facilitates the network to focus on valuable information in the sequences and ignore irrelevant information during its learning. We have conducted experiments on two sequence classification tasks for pattern classification and sentiment analysis, respectively. The experimental results demonstrate the advantages of AAMM over LSTM and some other related approaches. Hence, AAMM can be considered as a substitute of LSTM in the sequence learning applications.

Keywords Machine learning · Sequence classification · Attention mechanism · Long short-term memory (LSTM) · Attention-augmented machine memory (AAMM)

Introduction

The attention mechanism plays a critical role for humans in daily life. With the attention mechanism, humans can automatically notice the important information and neglect unimportant information during the perceptive and cognitive processes [1–3].

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In recent years, many cognitive computation models try to apply the attention mechanism and obtain promising learning results [4]. Among others, many attention-based deep neural networks have been developed [3, 5–7], and achieved good performances in some applications, such as neural machine translation [8] and image caption generation [3].

In order to learn effective features from spatial or temporal sequences, many machine memory models with the ability to memorize sequential information are widely applied. For example, recurrent neural network (RNN) is one of the most widely used machine memory model, which can theoretically memorize either long-term or short-term sequential information. Following RNN, many memory machines have been designed [9, 10]. In particular, the Long Short-Term Memory (LSTM) network is a special model of RNN and also a classical machine memory model with excellent performance in sequence learning [10]. Nevertheless, LSTM has no attention mechanism to pay attention to important information in the sequences. To address this problem, many researchers try to apply the attention mechanism to LSTM [3, 8, 11]. For instance, in [11], the researchers have proposed a self-attention mechanism and added it on a bidirectional LSTM. However, most of the existing attention-based LSTM models only leverage the attention mechanism outside the LSTM cells and have not thoroughly tackled the problem that LSTM lacks the attention mechanism.

In this paper, we present a novel model called attention-augmented machine memory (AAMM), which seamlessly incorporates the attention mechanism into the LSTM cell. Concretely, in AAMM, we embed an attention gate into the memory cell of LSTM. As a result, more than processing long short-term sequences, AAMM can focus on important and valuable information in the sequences. The attention gate works on the forget gate and the input gate. To some extent, it determines which part of information should be ignored and which part of information AAMM should pay attention to. To the end, the attention gate maintains the learning process of AAMM focusing on important information and neglecting irrelevant information. Functionally, AAMM can not only selectively remember critical historical information but notice more noteworthy information during the sequence learning. Compared to previous attention-based models, AAMM can be distinguished from them by its novel architecture and efficient implementation. The main contributions of our work can be summarized as follows:

- We design the attention gate that accepts inputs from the input gate and forget gate of AAMM. Specifically, the attention gate determines the attention distribution on the sequential information.
- We seamlessly integrate the attention gate into the memory cell of AAMM, which enables AAMM to focus on important parts of the input data and automatically ignore the irrelevant parts.
- Experiments on four standard benchmarks demonstrate that AAMM performs better than LSTM and other related models.

This paper is an extension to our conference paper with substantial improvements [12]. We have updated the proposed model and conducted more experiments for comparison with state-of-the-art approaches. The rest of this paper is organized as follows. In Section 2, we introduce some related work, including the attention mechanism and LSTM. In Section 3, we present the proposed AAMM model in detail. Section 4 reports the experimental results on two tasks, i.e., pattern classification and sentiment analysis, with comparison to LSTM and some other related models. Section 5 concludes this paper with remarks. Here, we emphasize that our motivation is to alleviate the issue that LSTM has no attention mechanism. Hence, the closely related and direct baseline method of AAMM is LSTM. Similar with previous attention-based LSTM models, we can also add the "attention" outside the cells of AAMM, but this is not in the scope of our research.

Related Work

In this section, we mainly introduce some related work from two aspects, the attention mechanism and the LSTM model.

The Attention Mechanism

The primary function of the attention mechanism is selection and allocation [2]. It leads to quick processing of information, with an efficient information choice and concentration of the computing power on the crucial tasks [2]. [1] introduces the attention mechanism in the human cognitive system, with which human pays attention to the important information and ignores unimportant information [1, 3]. In the cognitive computation area, the attention mechanism has been widely applied. In particular, a large amount of attention-based deep learning models have been proposed in recent years, and used to solve many real-world applications, such as object recognition, image caption generation and natural language processing [3, 13, 14].

In the literature, the applications of the attention mechanism can be divided into two categories: sole attention and model attention. The sole-attention models only utilize the attention mechanism without any recurrent or convolutional architecture, while the model-attention models incorporate the attention mechanism with other models. In the following, we introduce the models of these two categories, respectively.

Sole Attention In [7], the self-attention mechanism is used to replace the recurrent and convolutional layers that are commonly used in encoder–decoder architectures. It draws global dependencies between the input and output and leads to significantly faster training speed than architectures based on recurrent or convolutional layers. Moreover, the directional self-attention network (DiSAN) completely relies on the attention mechanism without any structure of RNN and convolutional neural network (CNN) [6]. It is specifically used to the sentence encoding applications. In DiSAN, the input sequence is processed by directional (forward and backward) self-attentions to model context dependency and produce context-aware representations.

Model Attention The memory network [15, 16] is a type of model attention, which introduces extra external memory into the neural network. In [15], a neural network with a recurrent attention model is presented. It is trained in the end-to-end manner, and requires significantly less supervision during training than previous memory networks, making it more generally applicable in realistic settings. Furthermore, [17] introduces the structured attention networks, which incorporate graphical models to generalize simple attention models. [18] presents a unified hierarchical attention model combining the strength of extractive and abstractive summarization, and introduces a novel inconsistency loss function to balance the two levels of attention. In addition, [19] constructs a simple but accurate content selector using the bottom-up attention, which restricts the ability of abstractive summarizers to copy words from the source and improves the model performance.

The models mentioned above are mainly multiplicative, i.e., giving different weights to the items. In this paper, we present a form of *additive* attention, i.e., adding different value to the items and adjusting them. Particularly, it is integrated in the memory cell of LSTM. Hence, it is quite different from the previous attention models.

LSTM

LSTM is a widely used deep learning model for sequential data [10]. As a special RNN model, it effectively alleviates the gradient exploding and gradient vanishing problems when dealing with long sequences. Till now, it has been applied to many sequence learning tasks, such as natural language translation [20–23], semantic recognition [24, 25], image and video description [26], and sentiment analysis [27].

Figure 1 shows the structure of the memory cell of LSTM, which includes three gates: the input gate, the output gate, and the forget gate.

In the learning of LSTM, the state of its memory cell is updated repeatedly over time [28]. The combined effect of the three gates is employed to process long short-term information.

The general updating formulas of LSTM are as follows:

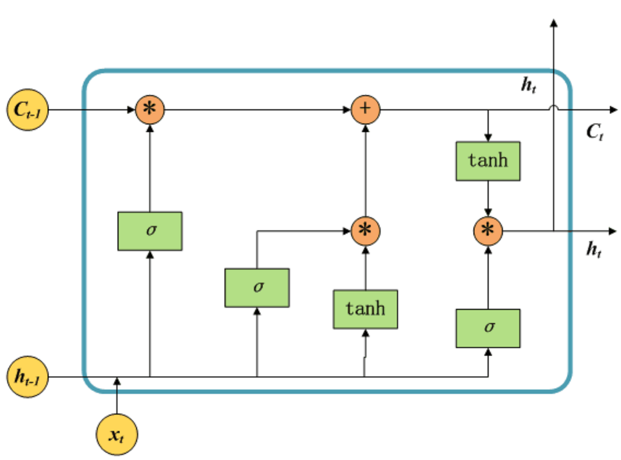


Fig. 1 The structure of the memory cell of LSTM

$$
\begin{aligned}
f_t &= \sigma(U_f[h_{t-1}, x_t] + b_f),\\
i_t &= \sigma(U_i[h_{t-1}, x_t] + b_i),\\
\tilde{C}_t &= \tanh(U_{\tilde{c}}[h_{t-1}, x_t] + b_{\tilde{c}}),\\
C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t,\\
o_t &= \sigma(U_o[h_{t-1}, x_t] + b_o),\\
h_t &= o_t * \tanh(C_t).
\end{aligned} \tag{1}
$$

In the above formulas, the weight parameters are denoted as $U_f, U_i, U_{\tilde{c}}, U_o$, and the biases are denoted as $b_f, b_i, b_{\tilde{c}}, b_o$. The forget gate f_t performs the function of forgetting information based on the current input datum x_t and the output of the previous cell h_{t-1}. Similarly, the input gate i_t and the output gate o_t determine what information to be added and output based on x_t and h_{t-1}. Here, σ denotes the sigmoid function. Besides, the candidate value $\tilde{C}_t$ is obtained based on x_t and h_{t-1} after the tanh activation. The information entered into the cell is calculated by element-wise multiplication between i_t and $\tilde{C}_t$. Analogously, the elements multiplication between f_t and the state of the cell at the last step C_{t-1} can generate the forgotten information. Then, the current state of the LSTM cell C_t is the sum of the above two terms. It gets the new state of the current LSTM cell updated by the input and forgotten information. After that, the current cell output h_t is obtained by o_t and C_t, which is activated by the tanh function.

To improve the learning ability of LSTM, many extensions of it have been proposed [25, 29]. In [29], the tensorized LSTM is proposed to increase the capacity of LSTM by widening and adding layers. For concreteness, the hidden states are represented by tensors and updated via a cross-layer convolution. By increasing the tensor size, the network can be widened efficiently without additional parameters; by delaying the output, the network can be deepened implicitly with little additional runtime since deep computations for each time step are merged into temporal computations of the sequence. The spatiotemporal LSTM (ST-LSTM) of [28] is an architecture for predictive learning, which enhances LSTM by memorizing spatiotemporal information. ST-LSTM models spatial and temporal representations in the memory cell and transmits the memory both vertically across layers and horizontally over states. In [30], the phased LSTM extends LSTM by adding a new time gate. This gate is controlled by a parametrized oscillation with a frequency range. By extending the fully connected LSTM to have convolutional structures in both the input-to-state and state-to-state transitions, [31] proposes the convolutional LSTM (ConvLSTM), which is an end-to-end trainable model. However, as discussed above, LSTM model and most of its extensions mainly focus on processing the sequential data, but cannot pay more attention to the important information in the sequences.

In order to address the above problem, some approaches try to combine the attention mechanism and the LSTM model together. For example, Liu et al. propose a global context-aware attention LSTM (GCA-LSTM) model for 3D motion recognition [32]. With the attention mechanism, GCA-LSTM can selectively focus on information nodes in action sequences according to the global context information. Moreover, Gao et al. present an end-to-end framework, attention-based LSTM, which adds the attention mechanism to the outside of the LSTM cells and can capture the significant structure of the video and translating it into descriptive statements [33]. In [34], to tackle the multivariate time series prediction tasks, an evolutionary attention-based LSTM (EA-LSTM) model is proposed, which applies an evolutionary attention learning approach outside the memory cell of LSTM. In [35], with the aim to explore the potential of the combination of attention clusters for multi-modal integration, the authors present a local feature integration framework based on attention clusters to account for the characteristics of the learned features and capture diverse signals in video classification task. Some other models that combine the attention mechanism with LSTM may include the attention-based hybrid LSTM-CNN model[36], the attention-based bidirectional LSTM (AB-LSTM) model [37] and the attention-based LSTM model proposed in [38]. However, these models just apply the attention mechanism to the outputs of the LSTM cell, that is, they do not seamlessly integrate the attention mechanism into the memory cell of LSTM, which is quite distinct from the proposed AAMM model in this paper.

Attention-Augmented Machine Memory

In order to seamlessly integrate the attention mechanism into the LSTM cell, we embed an attention gate in the memory cell of LSTM. For clarity, in Section 3.1, we introduce the attention gate first. And then, we describe the proposed AAMM model, which can be considered an extension of LSTM with the attention mechanism, in Section 3.2.

Attention Gate

In previous work, various attention mechanisms have been applied outside of the LSTM cells. The attention is generally computed and normalized with the *softmax* function, and multiplied with the outputs of the cells. Hence, it is "multiplicative". Here, we introduce an "additive" attention mechanism, which is incorporated into LSTM by seamlessly embedding an attention gate in the memory cell of LSTM. Fig. 2 shows the architecture of the attention gate.

Note that the attention gate is built upon the forget gate and the input gate. In this case, the attention gate can determine which part of information can be ignored and which part of information should be noticed. Hence, it is beneficial to not only memorize important information but also remove irrelevant information during the sequence learning. To update the attention gate, we need compute

$$\tilde{A}_t = \tanh(U_{\tilde{a}}[f_t, i_t] + b_{\tilde{a}}), \tag{2}$$

$$\hat{A}_t = \sigma(U_{\hat{a}}[f_t, i_t] + b_{\hat{a}}). \tag{3}$$

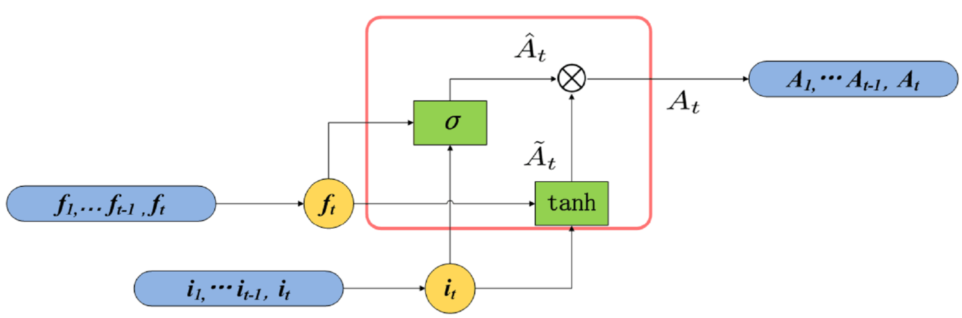
Here, the weight matrices are denoted as $U_{\tilde{a}}, U_{\hat{a}}$ and the bias vectors are denoted as $b_{\tilde{a}}, b_{\hat{a}}$. Equation 2 uses the *tanh* function to learn how much attention should be paid to the information from the forget gate and the input gate, respectively, while Eq. 3 applies the *sigmoid* function σ to determine the proportion to increase or decrease the attention.

The attention gate can be updated as

$$\begin{aligned} A_t &= \psi(\hat{A}_t[f_t, i_t], \tilde{A}_t[f_t, i_t]) \\ &= \hat{A}_t \otimes \tilde{A}_t, \end{aligned} \tag{4}$$

where $\psi(\mathbf{x}, \mathbf{y}) = \mathbf{x} \otimes \mathbf{y}$ is the element-wise multiplication operation. Elements in the candidate attention values $\tilde{A}_t$ may be positive or negative, which can be regarded as the initial attention distribution. Among them, positive elements represent valuable information and negative elements represent irrelevant information. The multiplication between $\tilde{A}_t$ and $\hat{A}_t$ gives the output of the attention gate A_t, which adjusts the initial attention distribution. In this case, A_t selectively focuses on important information in the sequences, by

Fig. 2 The architecture of the attention gate



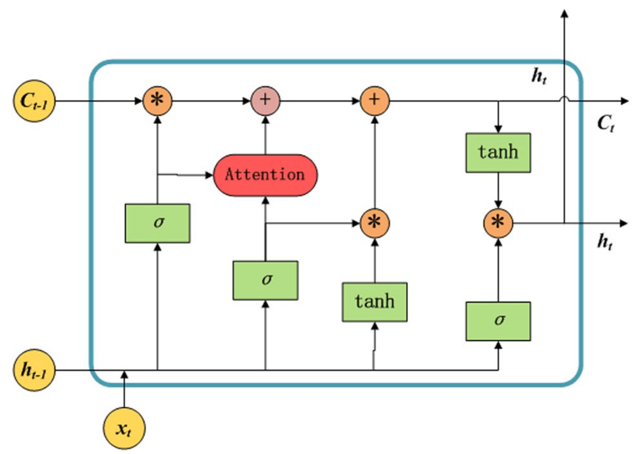


Fig. 3 The memory cell of AAMM. The highlighted part with red color is the attention gate

accepting useful inputs and removing irrelevant historical information.

AAMM

Based on the attention gate introduced in the last subsection, we present the proposed AAMM model in this subsection. Figure 3 shows the architecture of the memory cell of AAMM. We can see that AAMM inherits the forget gate and the input gate of LSTM, seamlessly integrates the attention gate and correspondingly updates the output gate. To the end, the attention mechanism is incorporated in LSTM. Particularly, comparing to the architecture of the memory cell of LSTM as shown in Fig. 1, we can see that the difference between LSTM and AAMM is obvious: AAMM adds an attention gate in the memory cell. That is, AAMM has the attention mechanism, but LSTM lacks it. Naturally, the attention mechanism is applied in the learning process of AAMM. Moreover, the attention mechanism of AAMM is distinct from previous attention-based LSTM extensions, where the attention is added after the whole sequence is dealt with by all the unfolded memory cells. Hence, AAMM is quite different from LSTM and its previous attention-based extensions (see the models introduced in Section 2.2). It is an attention-centered RNN model. During the sequence learning, it can adaptively focus on important information and ignore unimportant information.

For the updating of the forget gate, the input gate and the attention gate of AAMM, we can use the formulas given in Eqs. (1) and (4). The cell state can be updated as follows:

$$\begin{aligned}\hat{C}_t &= C_t + A_t \\ &= f_t * C_{t-1} + i_t * \tilde{C}_t + A_t \\ &= f_t * C_{t-1} + i_t * \tilde{C}_t + \tilde{A}_t \otimes \hat{A}_t.\end{aligned} \tag{5}$$

Please note that A_t is the output of the attention gate, such that the attention mechanism is additive, which adjusts information by additionally increasing or decreasing the element values of the cell state. Therefore, a well-trained AAMM cell can automatically adjust itself via selectively focusing on important input information and removing irrelevant historical information. It is distinct from previous attention mechanisms.

With the attention gate, AAMM can not only process the memory information, but also deliver effective outputs. Accordingly, its output gate can be updated as

$$h_t = o_t * \tanh(\hat{C}_t). \tag{6}$$

This formula is similar with that shown in Eq. (1) for the updating of the output gate of LSTM. However, its computing is based on the state of the memory cell of AAMM. Hence, it conveys relatively more important information to the next cell than that in the original LSTM.

Similar with LSTM, AAMM is a differentiable architecture and can be optimized easily. The difference is that AAMM applies the attention mechanism during its learning, so that it is able to process sequences by paying attention to important information and ignoring irrelevant information in the sequences. Moreover, AAMM can be utilized to any scenario that LSTM can be applied to. Therefore, it can be considered as a substitute of LSTM in the sequence learning applications.

Experiments

To evaluate the proposed AAMM method, we conducted extensive experiments on the pattern classification (Section 4.1) and sentiment analysis (Section 4.2) tasks, respectively. Particularly, we set LSTM as the baseline, as it is the closest model to AAMM. We implemented our experiments based on the TensorFlow platform. For LSTM, we downloaded the code from GitHub and directly applied the provided hyper-parameters. In order to fairly compare AAMM and LSTM, except for the attention gate, we employed same hyper-parameters for them. Specifically, ADAM was used as the optimization algorithm. Furthermore, we also compared some other state-of-the-art models, including some extensions of LSTM and several related approaches. In the following, we report the experimental settings and results.

Experiments on Pattern Classification

Pattern classification is a classical machine learning task, which is intensively researched in recent years. It includes handwritten digits classification, object recognition and so forth. In order to test AAMM on pattern classification applications, we mainly used the MNIST and Fashion-MNIST data sets. MNIST is a handwritten digits data set that can be used to evaluate the performance of the proposed model on handwritten digits classification tasks. It contains 70,000 gray scale images with size 28×28. The digits belong to 10 classes corresponding to 0-9. It has 60,000 images in the training set and 10,000 in the test set [39]. Alternatively, Fashion-MNIST is an image data set, whose number of data, number of classes, image format and partition are the same as MNIST [40]. Fashion-MNIST data set can be utilized to test the proposed model's performance on the object recognition task. In particular, in our experiments, we consider *the rows of an image in both data sets as sequential data* for sequence classification.

To analyze the performance of AAMM, we conducted experiments from two aspects. On the one hand, we compared it with some related RNN models, including gated recurrent unit (GRU) [9], LSTM [10], bidirectional LSTM (Bi-LSTM) [41] and nested LSTM (NLSTM) [42]. On the other hand, to analyze the advantages of AAMM over LSTM, we performed ablation study to show the effect of the attention mechanism.

Comparison with Related Models

Table 1 shows the classification results obtained by AAMM and the compared models, i.e., GRU, LSTM, Bi-LSTM and NLSTM, on the MNIST and Fashion-MNIST data sets. We can see that AAMM consistently outperforms all the compared models on the two used data sets. This shows the advantages of AAMM over previous RNN models on sequence classification tasks.

Ablation Study

AAMM integrates the attention mechanism into the memory cell of LSTM to improve its performance. In order to test the effect of the attention mechanism, we conducted ablation study to compare AAMM and LSTM on the Fashion-MNIST data set.

Figure 4 shows the curves of the loss function against the training steps obtained by LSTM and AAMM on the Fashion-MNIST data set. Note that LSTM and AAMM used the same loss function in our experiment. As we can see from Fig. 4 that, AAMM converges much faster than LSTM and has less vibration during its learning process. This indicates that with the attention mechanism, AAMM performs more efficiently and robustly than LSTM on the classification of the sequential data.

Table 1 The accuracy obtained by AAMM and related models on the MNIST and Fashion-MNIST data sets

Data set	GRU	LSTM	Bi-LSTM	NLSTM	AAMM
MNIST	97.79	97.47	97.81	97.75	97.85
Fashion-MNIST	88.16	87.46	88.18	88.32	88.60

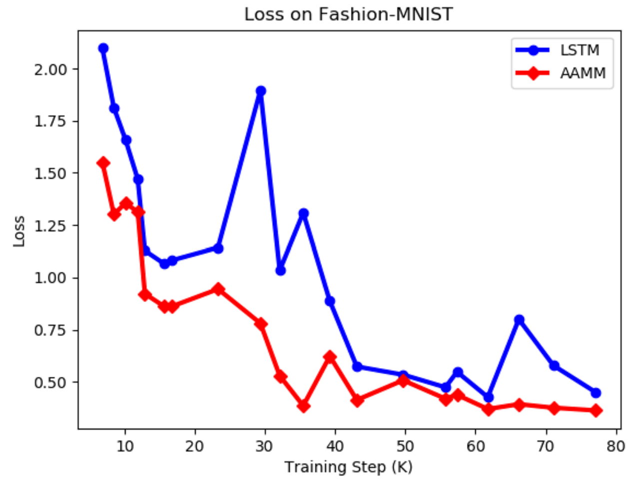


Fig. 4 The values of the loss function against the training steps obtained by LSTM and AAMM on the Fashion-MNIST data set

Table 2 shows the test accuracy obtained by LSTM and AAMM on the Fashion-MNIST data set at the 20,000th, 40,000th and 60,000th training steps. We can easily see that, AAMM outperforms LSTM consistently. In particular, in the early training steps (e.g., at the 20,000th step), AAMM performs much better than LSTM with the assistance of the attention mechanism. Since LSTM and AAMM can memorize the sequence data, their performances may be gradually close to each other as the iteration increases (e.g., with 60,000th steps). However, AAMM can outperform LSTM by more than 1%. This is similar to the human perception, where learning effect is better if we pay attention than not to do that, especially during the early stages.

In Fig. 5, we show that accuracy trend lines of LSTM and AAMM based on the results provided in Table 2. We can see that the accuracy trend line of AAMM is consistently above that of LSTM, which indicates that AAMM consistently outperforms LSTM on the learning tasks. Moreover, the slope of the accuracy trend line of AAMM is less than that of LSTM. Hence, AAMM is more stable than LSTM, which means that AAMM can obtain much better classification results with few iterations than LSTM.

To the end, we can conclude that AAMM can perform better than LSTM with higher accuracy, faster convergence and more stable learning process. Although their performance may be close with long time training, AAMM still outperforms LSTM. Particularly, with the attention mechanism, AAMM performs much better than LSTM in the early learning stages.

Experiments on Sentiment Analysis

Emotions play a key role in interpersonal communication and decision-making processes, and the application of sentiment analysis in many fields is critical [43, 44], such as

Table 2 Test accuracy obtained by LSTM and AAMM on the Fashion-MNIST data set at different training steps (the unit is thousands of steps)

	Acc.(%)	Steps(K)	Acc.(%)	Steps(K)	Acc.(%)	Steps(K)
LSTM	46.93	20	81.40	40	87.46	60
AAMM	62.20	20	86.62	40	88.60	60

company strategies, marketing campaigns and product preferences. Sentiment analysis is also an important and challenging machine learning task [45–49]. In order to evaluate the performance of AAMM on sentiment analysis, we conducted experiments on both the classical sentiment analysis and aspect-based sentiment analysis tasks.

Experiments on Classical Sentiment Analysis

To test the performance of AAMM on classical sentiment analysis, we used the internet movie review database (IMDB) [48]. IMDB is a crawler data set about the internet movie reviews. It divides all the film reviews into either the positive or the negative category based on the emotion of the reviews. This data set consists of 50,000 movie reviews. We used 25,000 reviews for training and the rest for test, with the same protocol proposed by [48].

In this experiment, we compared AAMM with LSTM and the hybrid deep belief network (HDBN) [50], both of which are classic models used for sentiment analysis. Figure 6 shows the error rates obtained by LSTM, HDBN and AAMM, while Fig. 6 shows the average running time per epoch of AAMM and the compared methods. As we can see, AAMM achieves the best classification result and uses the least running time per epoch among the compared methods. This demonstrates the effectiveness of AAMM on classic sentiment analysis. Particularly, comparing to LSTM, it can be seen that the attention mechanism benefits the learning of AAMM on this task.

Experiments on Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis is one of the most challenging tasks for sentiment analysis [51, 52]. In order to perform the aspect-based sentiment analysis and verify the effectiveness of AAMM, we conducted experiments on two data sets. One is a competition data set, i.e., SemEval 2014 Task 4 (SemEval14) [49], which contains two domains (Restaurant and Laptop). The other is a Twitter data set, collected by Dong et al. [47]. In these two data sets, the aspect terms of each review are labeled by three sentiment polarities, which are positive, neutral and negative, respectively. For example, about an aspect term *fajitas*, when it is in a sentence *"I loved their fajitas, but the service is horrible"*, its polarity is positive, but for aspect term *service*, its polarity is negative. Concretely, the statistics of the two data sets are given in Table 3.

In this experiment, we used the accuracy (Acc.) and F_1 measure (F_1) [53] as the measurements to evaluate the performance of AAMM and the compared methods. The reason for that we use F_1 as measurement is because the data sets have unbalanced classes as shown in Table 3 [47, 54], and it considers the class size in each data set. The F_1 can be computed as

$$F_1 = \frac{2 \times P \times R}{P + R}, \tag{7}$$

where "P" denotes the precision and "R" denotes the recall.

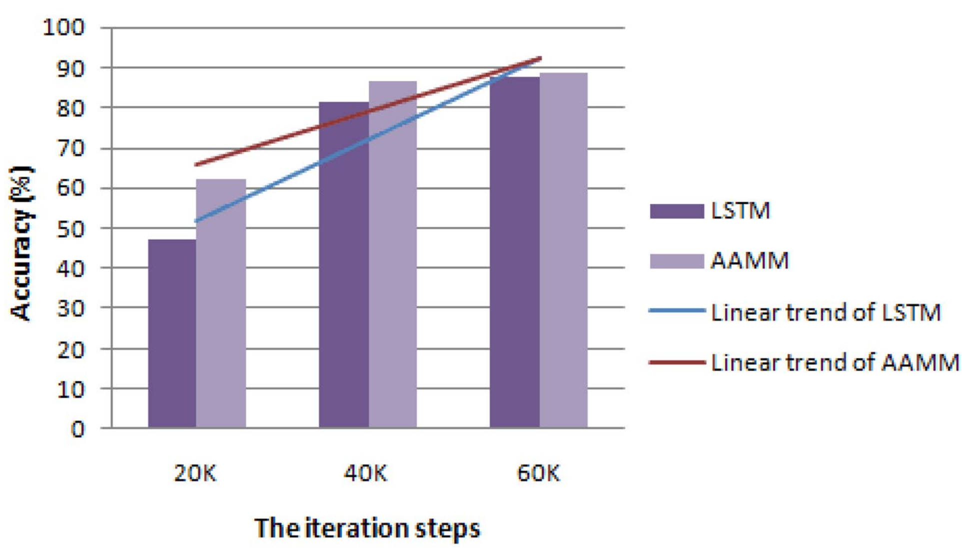
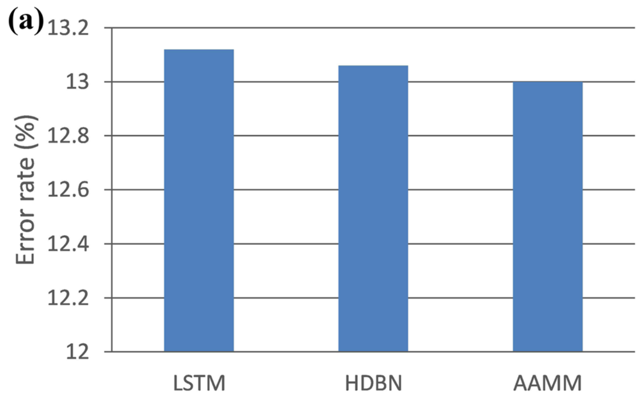


Fig. 5 Accuracy trend analysis of LSTM and AAMM on the Fashion-MNIST data set



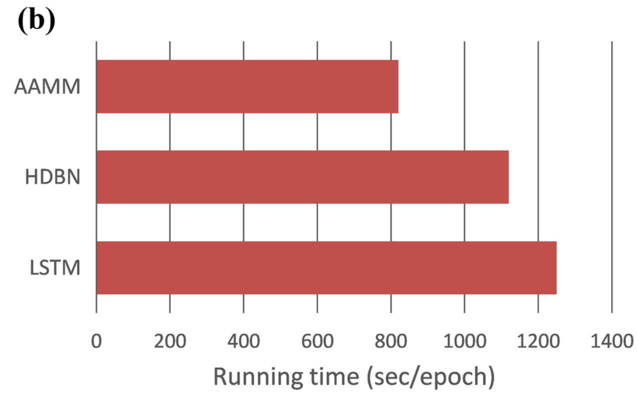


Fig. 6 Comparison between AAMM and related methods on the IMDB data set. Here, HDBN is for short of hybrid deep belief network [50]. **(a)** shows the error rates obtained by LSTM, HDBN and AAMM. **(b)** shows the average running time per epoch of LSTM, HDBN and AAMM, respectively

To show the advantages of AAMM over previous methods for aspect-based sentiment analysis, we compare it with the following methods:

- **Majority**: It is a basic method, which assigns the majority sentiment polarity labels in the training set to each sample in the test set [54–56].
- **AverageContext**: There are two versions of this method. The first one, named **AC − S**, averages the word vectors before the opinion target and that after the opinion target. The second one, named **AC**, averages the word vectors of the full context [57].
- **ATAE − LSTM**: It is an attention-based LSTM model with aspect embedding for aspect-based sentiment analysis [58]. It can pay attention to the specific parts of a sentence when several aspects are taken as input.
- **Cabasc**: It is a content attention model for aspect-based sentiment analysis. This model utilizes two attention enhancing mechanisms: the sentence-level content attention mechanism and the context attention mechanism [51].
- **MemNet**: It applies the attention mechanism multiple times on the word embeddings, and learns the attention weights on the context word vectors [54].

Table 4 shows the results obtained by AAMM and other compared methods. As can be seen, AAMM achieves the best results with respect to both the measurements, accuracy and F_1 measure. This demonstrate the advantages of AAMM over previous methods for aspect-based sentiment analysis, including LSTM and its attention-based variants.

Table 3 The statistics of the SemEval14 and Twitter data sets

Data set	Positive		Neutral		Negative	
	Train	Test	Train	Test	Train	Test
SemEval14(Restaurant)	2164	728	637	196	807	196
SemEval14(Laptop)	994	341	464	169	870	128
Twitter	1561	173	3127	346	1560	173

Table 4 The results obtained by AAMM and the compared methods on the SemEval14 and Twitter data sets. Concretely, the results of Majority, AC, AC-S, and MemNet were taken from the corresponding references, while that of the rest methods were obtained based on our implementation of the methods

Method	SemEval14 (Restaurant)		SemEval14 (Laptop)		Twitter	
	Acc.(%)	F_1	Acc.(%)	F_1	Acc.(%)	F_1
Majority [55]	65.00	0.3333	53.50	0.3333	50.00	0.3333
AC [57]	75.04	0.6396	67.29	0.6186	62.28	0.5912
AC-S [57]	75.85	0.6379	68.39	0.6217	63.29	0.6009
LSTM	77.41	0.6686	69.74	0.6394	68.93	0.6699
ATAE-LSTM	77.86	0.6718	69.75	0.6425	69.65	0.6762
Cabasc	78.12	0.6743	70.84	0.6552	69.51	0.6707
MemNet [54]	78.16	0.6583	70.33	0.6409	68.50	0.6691
AAMM	78.57	0.6801	71.16	0.6559	69.94	0.6911

Conclusion

Attention is an important cognition process of humans. However, many machine learning models can only memorize the data information, but lack the attention mechanism to focus on informative part of data. In this paper, we integrate the attention mechanism into the LSTM cell, and propose the AAMM model. The key idea of AAMM is to seamlessly embed an attention gate into the cell of LSTM. To the end, AAMM can pay attention to valuable information in the sequences during its learning. Extensive experiments on sequence classification demonstrate the advantages of AAMM over LSTM, the attention-based extensions of LSTM and other state-of-the-art methods. Hence, AAMM can be considered as a substitute of LSTM in the sequence learning applications. For future work, we plan to optimize the structure of the attention gate and add theoretical interpretation to AAMM.

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Declarations

Ethical Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Conflicts of Interest The authors declare that they have no conflict of interest.

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